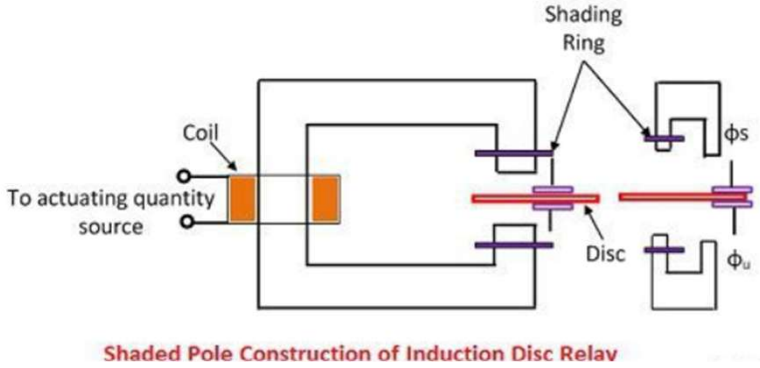
a. Shaded pole structure

-**Coil Energization:**
- Utilizes current flowing through a single coil.
- Wound on a magnetic structure featuring an air gap.

Air-gap Fluxes:
- Produced by the initializing current.
- Split into two fluxes that displace in time-space due to a shaded ring.

Shaded Ring:
- Composed of a copper ring encircling part of each pole face.



Shaded Pole Construction of Induction Disc Relay

Aluminum Disc Characteristics:
- Made of aluminum with low inertia.
- Requires less deflecting torque for movement.

Induction Mechanism:
- Two rings have current induced by the alternating electromagnetic flux.
- Magnetic field from the current causes flux in the iron ring to lag phase by 40° to 50° behind the flux in the unshaded portion of the pole.